

CHAIN ANALYTICS

Retail Supply Chain Operations Review

End-to-end analysis of orders, deliveries, returns and inventory

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Tools: Python, Pandas, NumPy, Seaborn, Matplotlib

github.com/HTS05/chain_analytics

1. Project Overview

This project began as part of a Chain Analytics Masterclass, which covered the basics of data cleaning on a single dataset. Building on that foundation, I extended the work into a full end-to-end retail operations analysis across five linked datasets — Orders, Products, Delivery, Returns and Inventory — covering delivery performance, revenue drivers, return patterns, store-level efficiency and stock health.

The workbook was built entirely in Python using Pandas for data wrangling and Seaborn/Matplotlib for visualization, following a structured analytics workflow: data quality checks → data modeling → KPI scorecard → deep-dive analysis → findings.

Objective

- Validate and merge five raw operational datasets into one clean, analysis-ready table.
- Build an executive KPI scorecard covering revenue, delivery performance, returns and inventory.
- Identify delivery bottlenecks, underperforming stores/partners and return drivers.
- Surface actionable insights for operations and inventory planning.

2. Dataset Overview

Five relational CSV files were used, joined on OrderID, ProductID and StoreID to build a single unified analysis table (final_united_dataset.csv).

Dataset	Rows	Columns	Duplicate Rows	Missing Cells
Orders	10,000	10	0	0
Products	51	6	0	0
Returns	1,000	9	0	0
Delivery	10,000	6	0	0
Inventory	5,100	6	0	0

Data Quality & Relationship Checks

- OrderID and ProductID confirmed unique across master tables.
- 100% of Delivery records and 100% of Return records matched back to a valid OrderID.
- Inventory is unique at the Store–Product level (no duplicate pairs).
- No negative quantities, revenue or refund values found — data passed integrity checks cleanly.

3. Methodology / Data Modeling

Key transformations applied before analysis:

- Converted OrderDateTime, ExpectedDeliveryTime, ActualDeliveryTime, ReturnDate and LastUpdated from text to proper datetime types.
- Computed delay_minutes (actual vs. expected delivery) and classified each order as Early, On Time or Delayed.
- Derived delivery_duration_minutes from each order's PickupTime and DeliveryEndTime timestamps.
- Aggregated returns to Order–Product level (return count, refund amount, reason) and merged into the master table.
- Took the latest inventory snapshot per Store–Product pair and flagged items at/below reorder level.
- Merged Orders + Products + Delivery + Returns + Inventory into one 10,000-row analysis table with zero unmatched records.

4. Executive KPI Scorecard

	KPI	Value
0	Total orders	10,000
1	Total revenue	₹5,014,167
2	Average order value	₹501.42
3	Average delivery time	16.4 minutes
4	On-time delivery	20.1%
5	Delay rate	79.9%
6	Return rate	10.0%
7	Refund value	₹479,203
8	Inventory value	₹448,286,264
9	Pairs at/below reorder	262

Insights Summary: *Recalibrate estimated delivery times* to fix the 79.9% delay rate, *enforce strict quality control* to stop the 10.0% return rate revenue leak (₹479K refunds), and automate purchase orders to restock the 262 low-stock product-store pairs.

5. Delivery Performance & Customer Impact

Delivery Status Breakdown

	Orders	Revenue	Avg_Delay_Minutes
delivery_status			
Early	0.0	0.0	0.000000
On Time	2010.0	1025791.0	0.000000
Delayed	7990.0	3988376.0	8.025782

Nearly 80% of all orders were delivered late, against expected delivery windows, making on-time delivery the single biggest operational gap in the business.

Returns by Reason

	Returns	Refund_Value
ReturnReason		
Stale Product	522	249275
Damaged Item	478	229928

Stale product complaints slightly outnumber damaged-item complaints, together accounting for the full ₹479K refund exposure — pointing to freshness/shelf-life and handling/packaging as the two core return drivers.

6. Store, Category, Product & Inventory Views

Top Stores by Revenue

	Orders	Revenue	AOV	Avg_Delivery_Minutes	Delay_Rate	Return_Rate	Reorder_Pairs
StoreID							
2016	125	71038	568.304000	16.888000	0.848000	0.120000	2
2063	111	67866	611.405405	16.891892	0.837838	0.135135	1
2079	110	67439	613.081818	16.718182	0.800000	0.081818	4
2002	108	65430	605.833333	15.824074	0.768519	0.101852	11
2066	115	64717	562.756522	17.347826	0.800000	0.113043	10

Store 2016 is the top revenue store, but it also carries one of the highest delay rates (84.8%) among leading stores — revenue leadership is not matched by delivery reliability there.

Revenue by Category

	Orders	Revenue	Quantity	Delay_Rate	Return_Rate
Category					
Grocery	2315	1481783	6935	0.794816	0.098488
Personal Care	2181	1253595	6480	0.797799	0.094452
Beverages	2178	1023266	6464	0.814509	0.100092
Snacks	1550	461932	4615	0.790968	0.103871

Grocery is the leading revenue category, followed closely by Personal Care and Beverages. Delay rates are fairly uniform (~79–82%) across categories, *meaning the delay problem is systemic rather than category-specific*.

7. Key Findings

- Delivery reliability is the biggest weakness: only 20.1% of orders arrived on time, with an average delay of 8 minutes on delayed orders.
- Store 2016 generates the most revenue (₹71,038) but also runs a high 84.8% delay rate — growth here is being undercut by service quality.
- Grocery is the top revenue category (₹1.48M), with Personal Care and Beverages close behind; delay rates are consistently high across all categories.
- Stale Product and Damaged Item are the two return reasons, together driving ₹479K in refunds — improving shelf-life management and delivery handling could meaningfully cut this.
- Inventory is largely healthy (95% of pairs), but 262 product-store combinations need reordering, worth about ₹3.2M in at-risk stock availability.

8. Root cause and Actionable Recommendation:

- **Delivery Reliability (20.1% On-Time)**
 - *Root Cause:* Promised delivery windows in the customer app are unrealistically tight given actual dispatch and transit times. Staffing shortages during off-peak hours (e.g., late-night shifts around 12 AM–1 AM) create severe fulfillment bottlenecks.
 - *Actionable Recommendation:* Recalibrate Estimated Delivery Times (EDTs) in the app to match real-world logistics capacity, and reallocate delivery driver shifts toward high-delay time slots.
- **High Revenue / High Delay (Store 2016)**
 - *Root Cause:* Order volume is outstripping store-level packing and dispatch capacity during peak sales hours.
 - *Actionable Recommendation:* Deploy dedicated order-packers during rush hours and set up a priority dispatch lane for top-performing revenue stores.
- **Product Returns (₹479K Refund Loss)**
 - *Root Cause:* Poor inventory batch rotation (FIFO failure) for perishables and insufficient protective packaging during transit.
 - *Actionable Recommendation:* Enforce strict First-In, First-Out (FIFO) stock guidelines, audit cold-chain maintenance, and upgrade fragile packaging.
- **Inventory Reorder Alerts (262 At-Risk Pairs)**
 - *Root Cause:* Delayed manual reorder triggers and slow supplier lead times.
 - *Actionable Recommendation:* Implement automated purchase order (PO) generation as soon as stock hits reorder thresholds, prioritizing high-volume locations.

8. Conclusion

The analysis shows a business with strong, stable revenue (₹5M+ across 10,000 orders) but a significant operational gap in delivery reliability, with delays affecting 4 out of 5 orders. Addressing delivery performance — especially at high-revenue but high-delay stores like 2016 — alongside tighter control on stale/damaged returns and proactive reordering, represents the clearest path to improving customer experience and protecting margin.

9. Project Links

GitHub Repository (notebook, dataset & certificate):

github.com/HTS05/chain_analytics

Certificate: Chain Analytics Masterclass — Heet Singha